" On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of those principles will lead to a zero grade and is subject to harsh discipline issues "

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1) a)

$$A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

They mentioned some of the elimination matrices so, to find the values of α, γ, β, k and m

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

$k = 14$

$\alpha^2 = 0 \Rightarrow \alpha = 0$, $\gamma = 0$

$\alpha = 0$

$\beta = -1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -\gamma & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

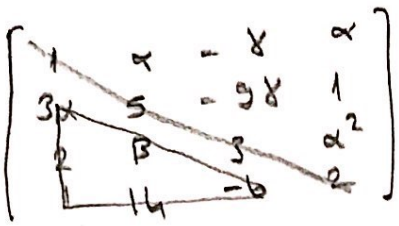
$m = 2$

$k = 14$, $\alpha = 0$, $\gamma = 0$, $\beta = -1$ $m = 2$

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b) Find the eliminated upper triangle matrice U=?

Upper Triangular matrise U. The elements below the dragonal are zero.



These elements are zero

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5 & -9\gamma & 1 \\ 0 & 0 & 3 & \alpha^2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

as α and $\gamma = 0$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Eliminated upper triangular matrise U

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

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3) a) Suppose, u is a unit vector in $\mathbb{R}^n$.

Hence, H is the $n \times n$ symmetric matrix

$$H = I - 2uu^T, \quad u \text{ is unit vector, so, } \|u\| = 1$$

$$H^2 = H.H = (I - 2uu^T)(I - 2uu^T)$$

$$= I^2 - I.2uu^T - 2uu^T I + 4uu^T uu^T$$

$$= I - 2uu^T - 2uu^T + 4u \|u\|^2 u^T$$

(As, $\|u\|^2 = u^T u$, $\|v\| = +\sqrt{u^T u}$

so, $H^2 = I - 4uu^T + 4uu^T$ $\Rightarrow \|u\| = 1$)

$$\Rightarrow \boxed{H^2 = I}$$

b) $$HH^T = (I - 2uu^T).(I - 2uu^T)^T$$

$$= (I - 2uu^T).(I^T - (2uu^T)^T) \quad [(A-B)^T = A^T.B^T]$$

$$= (I - 2uu^T).(I - 2(u^T)^T.u^T) \quad [(AB)^T = B^T.A^T]$$

$$= (I - 2uu^T).(I - 2uu^T) \quad (\text{As. } (u^T)^T = u)$$

$$= I^2 - 2uu^T - 2uu^T + 4uu^T uu^T$$

$$= I - 4uu^T + 4u\|u\|^2 u^T \quad (\|u\| = 1)$$

$$= I - 4uu^T + 4uu^T$$

$$= I$$

$H^T.H = I$, so $H.H^T = H^T H = I$

so, H is an orthogonal matrix

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c) For a matrix A, a vector (non zero) will be eigen-vector (suffose V) of A corresponding eigen value λ if, $\boxed{AV = \lambda V}$

Now, $Hu = (I - 2uu^T)\,u$

$= Iu - 2uu^Tu$

$= u - 2u\,\|u\|^2$

[As I is identity matrix, so 1 is the only eigen value of I and, $Iu = 1.u = u$ (also), and, $\|u\| = +\sqrt{u^Tu}$

u is unit vector, so, $\|u\| = 1 = +\sqrt{u^T.u}$ and hence, $u^Tu = \|u\|^2 = 1$]

$Hu = u - 2.u.1 = -u$

$\Rightarrow Hu = -u = (-1)u$

so, $Hu = \lambda u$, where, $\lambda = -1$

u is an eigen vector of H

corresponding eigen value for u is $= \lambda = -1$

d) $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T \quad (|A-B)^T = A^T - B^T)$

$= I - 2(u^T)^T u^T \quad (|AB)^T = B^T A^T)$

$= I - 2uu^T = H$

H is a symmetrix matrix

There is a theorem which states that,

$H = Q \Lambda Q^T$ (where, Q is an orthogonal matrix)

$= Q \Lambda Q^{-1} \quad (|QQ^T = I \Rightarrow Q^T = Q^{-1})$

$= (Q^{-1})^{-1} \Lambda Q^{-1} \quad (|\Lambda$ = Diagonal matrix comes to of eigen values of H)

$= P^{-1} \Lambda P$

So, H is diagonalizable matrix.

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4)

Given that

a) $\frac{dv}{du} \propto (v-w)$

$\frac{dv}{du} \propto -(v-w)$

$$\frac{d}{du}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & +1 \\ +1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

$v(0) = 10$
$w(0) = 40$

b) $\Rightarrow \frac{dv}{du} = (-v+w)$, $\frac{dw}{du} = v-w$

taking laplace of

$sV(s) - v(0) = -V(s) + W(s)$

$sW(s) - w(0) = V(s) - W(s)$

$\Rightarrow sV(s) + V(s) - W(s) = 10$

$sW(s) + W(s) - V(s) = 40$

$$\Rightarrow \begin{bmatrix} s+1 & -1 \\ -1 & s+1 \end{bmatrix}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} s+1 & -1 \\ -1 & s+1 \end{bmatrix}\begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$= \begin{bmatrix} V \\ W \end{bmatrix} = \frac{1}{[(s+1)^2-1]}\begin{bmatrix} s+1 & 1 \\ 1 & s+1 \end{bmatrix}\begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$= \begin{bmatrix} V \\ W \end{bmatrix} = \frac{1}{[(s+1)^2-1]}\begin{bmatrix} 10s+10+40 \\ 10+40s+40 \end{bmatrix}$$

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$$= \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} \frac{10s+50}{(s+1)^2-1} \\ \frac{40s+50}{(s+1)^2-1} \end{bmatrix}$$

$$= V(s) = \frac{10s+50}{s^2+2s}, \quad w(s) = \frac{40s+50}{s^2+2s}$$

taking inverse laplace

$$V(t) = 25 - 15e^{-2t}, \quad w(t) = 25 + 15e^{-2t}$$

Total mixtoso = $10+40 = 50$

at $t = 1$ tonk 1 $= 25 - 15e^{-2} = 22.969970$

$$\Rightarrow \% = \frac{22.969970}{50} \times 100 = 45.93\%$$

at steady state $\Rightarrow$ at $t \to \infty$

$V(t)$ $25 - 15e^{-\infty}$ (to 0) $\qquad w(t) = 25 + 15e^{-\infty}$ (to 0)

$\Rightarrow$ **25**

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